

Action-Mean GRPO: A Single-Step Baseline

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Abstract

This note specifies a single-step action-mean GRPO baseline for comparison with the two-step gated method. At each complete decision state, old visits are grouped by current action, terminal rewards are averaged within each action, and distinct action means are equally weighted in the advantage normalization. Every model-generated token in a step receives that step’s advantage. Tokenwise clipping and KL regularization use the same definitions and step/trajectory averaging as the two-step method. No implementation or experimental result is reported.

1 Relationship to the two-step method

One edit or tool call is one decision step. Both methods use complete old trajectories and terminal rewards, count repeated visits to the same complete state separately, and use the same action-mean normalization. They also share the generated-token scope, tokenwise probability ratios and clipping, tokenwise KL, and the same averaging order. Tool responses, environment observations, and prompts may condition generated tokens but are not policy prediction targets.

The baseline always uses the current step’s action-mean advantage. It does not select the bottom 20% of near-zero positions or add λ . It does not group successor actions a' , inspect child-state dispersion, require five distinct successor actions, apply the Δ gate, compute A_2 , use β , select second-step actions, or apply an overlap rule. These are differences of the two-step method, not omitted terms to be recovered elsewhere in this baseline.

2 Trajectories, states, and action groups

From the same tasks and initial conditions as the two-step method, collect N complete trajectories τ_i under the known old policy π_{old} . Trajectory i has T_i decision steps and terminal reward R_i . Let $s_{i,j}$ be its complete decision state at step j , and $a_{i,j}$ the edit or tool-call action. State equality must use the same rule in both methods. For each observed state s , set

$$\mathcal{I}(s) = \{(i, j) : s_{i,j} = s\}, \quad \mathcal{U}(s) = \{a_{i,j} : (i, j) \in \mathcal{I}(s)\}, \quad d_s = |\mathcal{U}(s)|. \quad (1)$$

For each observed action $u \in \mathcal{U}(s)$, average the terminal rewards over visits that took that action:

$$\mathcal{I}(s, u) = \{(i, j) \in \mathcal{I}(s) : a_{i,j} = u\}, \quad n_s(u) = |\mathcal{I}(s, u)|, \quad q_s(u) = \frac{1}{n_s(u)} \sum_{(i,j) \in \mathcal{I}(s,u)} R_i. \quad (2)$$

A trajectory that revisits exactly the same state contributes one record per visit. If it takes the same action more than once, its R_i appears more than once in the corresponding sum. Those visits are distinct decisions, not statistically independent trajectories.

3 Equal weighting of action means and step advantage

Give each of the d_s distinct observed actions equal weight when computing the mean and population standard deviation of action means:

$$\bar{q}_s = \frac{1}{d_s} \sum_{u \in \mathcal{U}(s)} q_s(u), \quad \sigma_{q,s} = \left[\frac{1}{d_s} \sum_{u \in \mathcal{U}(s)} (q_s(u) - \bar{q}_s)^2 \right]^{1/2}. \quad (3)$$

Assign the standardized value for the chosen action back to every visit:

$$A_{i,j}^{\text{base}} = A_{i,j}^{\text{state}} = \begin{cases} \frac{q_{s_{i,j}}(a_{i,j}) - \bar{q}_{s_{i,j}}}{\sigma_{q,s_{i,j}}}, & \sigma_{q,s_{i,j}} > 0, \\ 0, & \sigma_{q,s_{i,j}} = 0. \end{cases} \quad (4)$$

Visits with the same state and action share an advantage even when their own terminal rewards differ. One observed action or identical action means give zero dispersion and therefore zero advantage at that position; the trajectory itself is retained. Action means have equal weight in the normalization, but sampled visit positions are not reweighted in the policy objective by $1/n_s(u)$ or any other inverse action-frequency factor.

4 Token assignment and policy update

Let step (i, j) have $L_{i,j}$ model-generated tokens $y_{i,j,1:L_{i,j}}$, including its thinking text and edit or tool-call content. Every generated token receives the same step advantage:

$$A_{i,j,k}^{\text{base}} = A_{i,j}^{\text{base}}, \quad k = 1, \dots, L_{i,j}. \quad (5)$$

Let $h_{i,j,k}$ be the complete context before predicting token k . Compute a probability ratio and clipping for each generated token:

$$r_{i,j,k}(\theta) = \frac{\pi_{\theta}(y_{i,j,k} \mid h_{i,j,k})}{\pi_{\text{old}}(y_{i,j,k} \mid h_{i,j,k})}, \quad (6)$$

$$\ell_{\text{clip}}(r, A) = \min\{rA, \text{clip}(r, 1 - \epsilon, 1 + \epsilon)A\}. \quad (7)$$

The KL penalty is enabled at the same token scope, using the sampled-token estimator of GRPO [1]:

$$D_{i,j,k}^{\text{KL}}(\theta) = \frac{\pi_{\text{ref}}(y_{i,j,k} \mid h_{i,j,k})}{\pi_{\theta}(y_{i,j,k} \mid h_{i,j,k})} - \log \frac{\pi_{\text{ref}}(y_{i,j,k} \mid h_{i,j,k})}{\pi_{\theta}(y_{i,j,k} \mid h_{i,j,k})} - 1. \quad (8)$$

Average the clipped and KL terms over generated tokens within each step, then over steps within each trajectory, then over trajectories. With the same positive KL coefficient κ as the two-step method, maximize

$$J^{\text{base}}(\theta) = \mathbb{E}_{\{\tau_i\} \sim \pi_{\text{old}}} \left[\frac{1}{N} \sum_{i=1}^N \frac{1}{T_i} \sum_{j=1}^{T_i} \frac{1}{L_{i,j}} \sum_{k=1}^{L_{i,j}} \left(\ell_{\text{clip}}(r_{i,j,k}(\theta), A_{i,j}^{\text{base}}) - \kappa D_{i,j,k}^{\text{KL}}(\theta) \right) \right], \quad \kappa > 0. \quad (9)$$

Equivalently, minimize $-J^{\text{base}}(\theta)$. Advantages and action-group statistics are computed from old trajectories and held fixed during an update. Gradients pass through current-policy token ratios and tokenwise KL. Only model-generated tokens contribute to $L_{i,j}$ and the token sum. The numerical κ and ϵ , as well as the reference-policy choice, remain unspecified; they must match the two-step comparison.

5 Calculation and comparison boundary

Collect old trajectories with the same old policy, tasks, sampling budget, and terminal reward definition as the two-step method. At each complete state, group visits by their current action, compute $q_s(u)$, equally

standardize the distinct action means, and return $A_{i,j}^{\text{base}}$ to every step. Broadcast it to generated tokens, compute tokenwise ratios, clipping, and KL, then apply the step/trajectory averages and update the policy. Ignoring a' means no explicit grouping, selection, gate, or advantage adjustment based on a successor action or child state. Terminal reward still depends indirectly on later behavior; excluding all later outcomes would require a different reward definition and a different baseline. For a controlled comparison, the two methods should share terminal rewards, state matching, trajectory sampling, generated-token scope, clipping, KL definition and coefficient, loss averaging, and evaluation.

6 Open experimental settings

Both methods use tokenwise KL. Their common reference policy and numerical κ remain to be set, as do ϵ , data splits, sampling budget, failure handling, stopping conditions, and evaluation metrics. This note is an algorithm proposal, not a formal experimental configuration or a report of experimental results.

References

- [1] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. Deepseekmath: Pushing the limits of mathematical reasoning in open language models, 2024. arXiv:2402.03300.